Electromyography-Based Locomotion Pattern Recognition and Personal Positioning Toward Improved Context-Awareness Applications

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***Abstract*—Personal positioning has been playing an important role in context awareness and navigation. Pedestrian dead reckoning (PDR) solution is a positioning technology used where the global positioning system (GPS) signal is not available or its signal is mightily attenuated or reflected by constructions nearby, such as inside the buildings or in GPS degraded areas such as urban city, basement. A traditional PDR solution employs a multisensor unit (integrating accelerometer, gyroscope, digital compass, barometer, etc.) to detect step occurrences, as well as to estimate the stride length. In our pilot research, we proposed a novel electromyography (EMG)-based method to fulfill that task and obtained satisfying PDR results. In this paper, a further attempt is made to investigate the feasibility of using EMG sensors in sensing muscle activities to detect the corresponding locomotion patterns, and as a result, a new approach, which recognizes different locomotion patterns using EMG signals and constructs stride length models according to the recognition results, is then proposed to improve the positioning accuracy and robustness of the EMG-based PDR solution by adapting the stride length model into different locomotion patterns. The experimental results demonstrate that EMG-based pattern recognition of four motions (walking, running, walking upstairs, walking downstairs) achieve an error rate of less than 2%. Combined with locomotion pattern recognition, the proposed EMG-based PDR solution yield a position deviation of less than 5 m within the whole distance of 404 m in a simulated indoor/outdoor field test. The proposed method is proven to be effective and practical in sensing context information, including both the user's activities and locations.**

***Index Terms*—Electromyography (EMG), locomotion pattern recognition, pedestrian dead reckoning (PDR), wearable sensors.**

Manuscript received June 6, 2012; accepted November 10, 2012. Date of publication June 3, 2013; date of current version August 14, 2013. This work was supported in part by the Fundamental Research Funds for the Central Universities of China under Grant WK2100230005 and Grant WK2100230002, and the National Nature Science Foundation of China under Grant 61271138. This paper was recommended by Associate Editor J. Bass. *(Corresponding author: Xiang Chen.)*

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Digital Object Identifier 10.1109/TSMC.2013.2256857

I. Introduction

IN RECENT years, context awareness [1]–[4] has become a popular research topic due to the rapid development of mobile computing and advanced network technologies. In computer science, the context awareness refers to a technique that enables the computing devices to sense information about the circumstances under which they are able to operate and react accordingly. In order to interact with human intelligently, context awareness technology embedded in the device aims to design a smart environment to be capable of inferring human's intentions. For instance, a context-aware mobile phone may know that the user has sat down in a meeting room and conclude that the user is currently busy in a meeting. It thereby may reject any unimportant calls or inform the potential callers of such situation [1]. Mobile healthcare is also one of the important applications of context awareness techniques, which are capable of monitoring people with chronic diseases in a long-term period, as well as alerting their caregivers that immediate intervention is required [2], [3], [5].

Positioning information is a very basic and important category of context, and has been by far frequently used for most context-aware applications. The personal positioning technology with global positioning system (GPS) brings people a great convenience in retrieving their own locations when doing outdoor activities, and has become a standard feature of smartphones. The availability and accuracy of GPS stand-alone solution is still far to reach a ubiquitous solution, restricted by the physical contributes of the GPS radio frequency signal. Therefore, along with the prevalence of radio frequency identification devices (RFID), wireless local area network (WLAN), and ultrasound technologies, a variety of indoor navigation studies [6]–[12] investigated the applications of these communication technologies to develop personal positioning in public places (such as supermarkets, airports, office buildings, housing districts, etc.). The diagram of this paper's work is shown in Fig. 1. For instance, Mazuelas *et al.* [6] utilized the WLAN received signals from different predetermined sites to infer the receiver's position according to the relation between the receiving signals' strength and propagation distance. Koutsou *et al.* [7] utilized the prior settled RFID to transmit identifiable RF signal periodically, and calculated the test receiver's position by analyzing signal strength, which yielded a position accuracy of 3.25 m.

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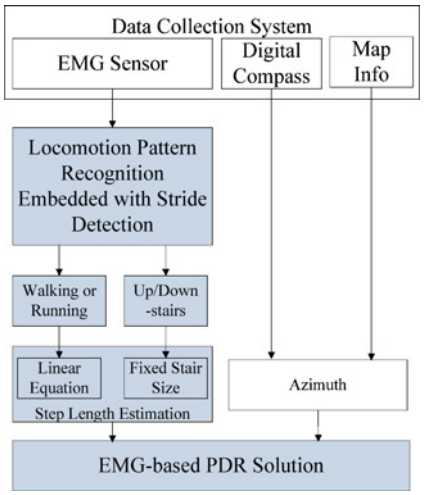


Fig. 1. Diagram of the proposed pedestrian dead reckoning solution.

Nevertheless, these methods mainly rely on the predefined infrastructures of the circumstance that restrict its applicability for massive applications. The positioning accuracy dramatically decreases when there are no or not enough such fixtures of this type. Recently, Ishikawa *et al.* [8] used positioning method based on RFID and cameras as an auxiliary of inertial measurement unit (IMU) to improve the accuracy of indoor personal positioning. IMU is a traditional tool used in personal positioning, including three-axi accelerometers and gyroscope. In pedestrian dead reckoning (PDR) based on IMU, position is calculated by using acceleration signals from an accelerometer settled on the back [9], [10] or waist [8], [13] for step detection, as well as step length estimation, and measurement from digital compass as heading information.

In addition, user behavior analysis is also an important feature of a context-aware system. Sensing user's behaviors usually relies on the ambient infrastructures (such as cameras [14], [15]), or wearable sensors [16]–[21], [41] attached to the user. Utilizing wearable sensors is preferred due to its mobility and has been prevalent in the research field of context awareness for years. For instance, Jonny *et al.* [16] utilized a sensor badge that integrated two accelerometers to identify the wearer's sitting, standing and lying, and they also made a sensor-integrated jacket to measure the wearer's (upper) arm-movements, which makes it possible for their wearable computers to gain the situational awareness. Hans *et al.* [1] proposed a method named technology enabling awareness to investigate the implementation of multisensor context awareness with a self-contained device and mobile telephones. Andreas *et al.* [17] introduced that they used eye motions as a new input modality for context awareness and mobile human–computer interaction (HCI) applications, and demonstrated that electrooculography is a suitable sensing technique for the recognition of reading activities.

Electromyography (EMG) is an electrophysiological technique that is capable of detecting the electrical activity produced in skeletal muscles during their contractions. Unlike the inertial sensors (e.g., accelerometer and gyroscopes) that measure some physical quantities of obvious locomotion, the EMG sensor directly measures electrical potentials generated by muscle contractions from the human body. The recorded EMG signals are able to reflect not only obvious human movements of the entire body or limbs in part, but also unapparent movement intensions. Therefore, EMG signals are widely applied in the fields of clinical gait analysis [22], prosthetic control [23], and gestural HCI [28], [29]. Specially, when we are walking, the muscles on both legs contract periodically according to each step, and the strength exerted by the leg has a positive correlation with the amplitude of the EMG signals. In this regard, sensing leg muscle activation via the EMG technique is a sufficient way in discriminating pedestrian's walking patterns, such as standing still, walking, and running fast or slow. In addition, such information can improve the positioning accuracy by some predefined knowledge [41].

With the above-mentioned consideration, in our early work [30]–[34], a novel EMG-based personal positioning method was presented, in which the walking steps were detected from the EMG signals recorded from both legs rather than accelerometers. EMG signals were also used for a PDR algorithm to determine the pedestrian's locations. Preliminary field tests showed satisfactory performance of the EMG-based PDR method in positioning with an error less than 5 m within a total walking distance of 270.8 m. Based on the comparison between accelerometer-based and EMG-based PDR, we found they have comparative positioning accuracy in field tests [31]. Despite the promising performance achieved by our preliminary efforts, the current EMG-based PDR method is, however, still restricted in the following aspects: 1) the currently employed stride length model cannot be adapted to various locomotion patterns involved in activities of daily life; 2) the heading information derived from a compass was very likely to be interfered by environmental disturbances as well as platform inclination; and 3) the proposed PDR solution is only appropriate for positioning in a 2-D environment.

The work described in this paper is evolved from our pilot studies, with the attempt to further investigate the feasibility of using EMG sensors in sensing muscle activities associated with locomotion patterns, thus improving the accuracy of PDR-based indoor positioning and extending it from 2-D to a 3-D environment. In this paper, a novel method, which combines the EMG-based PDR solution with locomotion pattern recognition, is proposed. In the proposed method, multichannel EMG signals recorded from lower limb muscles are utilized to identify pedestrian's activities, including walking, running, walking upstairs, and walking downstairs. A stride length model is developed for each locomotion pattern with respect to its kinesiological properties. Furthermore, considering the error of heading information derived from low-cost digital compass, map matching technology [35] is applied to reduce the position deviation contributed by heading errors.

The remainder of this paper is structured as follows. Section II describes the method in detail. Section III gives the field tests' results and discusses the efficiency of the proposed method in pedestrian dead reckoning, some relative issues and future work are discussed in Section IV. Section IV also concludes this paper.

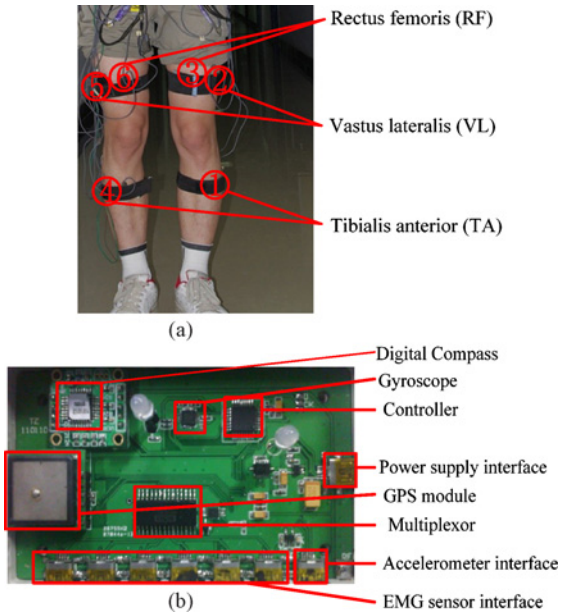


Fig. 2. (a) Six-channel EMG sensors are attached to TA, VL, RF of each leg. (b) Multisignal collection system.

II. Methods

A. Data Collection

1) *Wearable EMG Sensors and Setup:* Each EMG sensor used in the experiments was embedded with a pair of bar-shaped surface electrodes with 1 mm × 10 mm contact area and 10 mm interelectrode distance to allow differential recording in a channel. Each EMG sensor module, weighting less than 20 gram and measuring 38 mm × 20 mm × 10 mm, consisted of a built-in two-stage amplifier with a typical gain of 60 dB and a microcontroller with 12-b A/D converter. Multiple sensor modules could be distributed and placed over the skin surface of different muscles of interest to record multichannel EMG signals simultaneously.

Most locomotion tasks involve complex temporal and spatial coordination of multiple muscles. These muscles are responsible for the movements of ankle and knee joints. Based on the anatomic knowledge and some pretests prior to the experiment, it was found that the Rectus femoris (RF) and Vastus lateralis (VL) majorly accounted for the movement of knee joint, and Tibialis anterior (TA) was more likely to account for the movement of ankle joint. Therefore, three EMG sensors were placed on each leg targeting TA, RF and VL, respectively, as shown in Fig. 2(a). The sampling rate was 1000 Hz per channel and a representative signal segment was shown in Fig. 3. EMG recordings from different muscles were likely to display a similar cyclic pattern with respect to each step when the user is walking. In this paper, the signals recorded from VL and RF were selected for further process.

2) *Multisensor Positioning System With GPS Receiver:* In addition to EMG signals mentioned above, GPS and heading information were collected with a wearable multiple sensors positioning system developed by our group. The system [Fig. 2(b)] consisted of a multiplexor, a control unit, an onboard GPS module (Fastrax UP500, 10-Hz, Finland), a gyroscope (InvenSense ITG-3200, CA, USA), a two-axis digital compass (Honeywell HMC5883L, NJ, USA) and a group of extensible sensor sockets supporting up to six EMG sensors and one three-axis accelerometer. During data collection, the system was powered by a laptop computer via a USB cable that was also responsible for transferring experimental data into the computer. The software developed by the group installed in the computer was able to monitor all the recorded channels on the screen in real time. In this paper, the EMG data, heading information from the digital compass, and the GPS data were recorded in the laptop for further analysis.

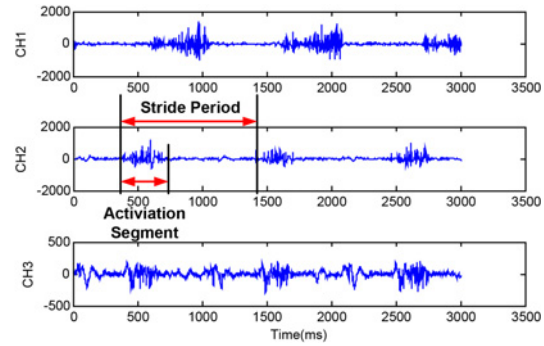
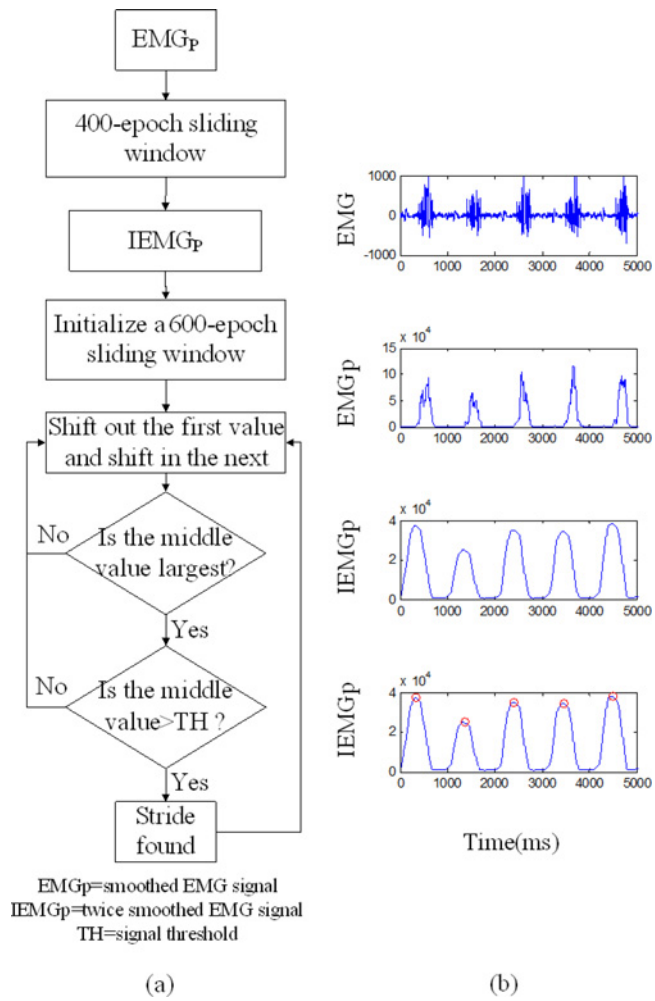


Fig. 3. Three-channel EMG signals of left leg.

B. Locomotion Pattern Recognition Embedded With Step Detection

1) *Walking and Non-Walking Activity Classification:* When walking, a stride period for one leg can be divided into stance phase and swing phase [36]. In stance phase, the foot exerts an impact on the ground, which leads to a large vibration period in the EMG signals recorded from the corresponding leg. Such a vibration period is viewed as an activation segment in this paper, as shown in Fig. 3. While in a swing phase, the foot is suspended to make a pace, so the leg muscles stay relaxed, and the EMG signals are more likely to show a relative quiescent baseline. Therefore, each step can be detected by finding its corresponding EMG activation segment within the signal sequence.

However, such an EMG activation segment does not always result from a walking step. There are a variety of such common situations in our everyday life. For example, when we transfer our weight from one leg to the other randomly in an elevator, or when we stop sharply after a long-time running, the muscles in legs can also contract and the corresponding EMG activation segments may be observed. To overcome the possible misdetection problem, a hidden Markov model (HMM) classifier was utilized to distinguish the walking pattern from other irregular movements in our preliminary work [30], taking advantage of its ability to identify the overall characteristics of the walking pattern that the muscles in left and right legs are alternatively activated to make each step. In [30], raw EMG signals were segmented into frames and sample entropy of every frame was extracted as the inputs to HMM. Although the results are satisfactory, the calculation of sample entropies takes too much time. In this paper, we use mean absolute value of each frame to replace sample entropy as the inputs of HMM.



(a) (b)

Fig. 4. (a) Flow diagram of step detection algorithm, the figures 400 and 600 are selected empirically. (b) CH2 EMG signal processing.

When the walking pattern had been identified, a simple and robust method for detecting step occurrence was performed on raw EMG signals, with an assumption that the HMMs classification result has given a real walking modality. The following approach performed on each leg for step occurrence detection and further locomotion pattern classification is universal for both legs. Therefore, the left side is set as an example for describing the proposed method.

2) *Step Detection Algorithm:* To eliminate environmental noise, we use a band-pass filter to eradicate signal with the frequencies outside 50–450 Hz that is the frequency domain of EMG signals. EMG signals collected from VL were used for step detection, which in our case are called channel 2 (CH2) for left leg and channel 5 (CH5) for right leg. A sliding mean square window was applied on the raw data to extract the EMG energy profile. The processed signal (denoted as EMG_P) can be calculated with

$$EMG_P(i) = \frac{1}{N}\sum_{i=j}^{i+N} EMG^2(j) \tag{1}$$

where $EMG(j)$ denotes the jth value of raw EMG signals, and $EMG_P(i)$ denotes the ith value of the processed energy profile. N is chosen depending on what occasions we are utilizing the algorithm in. The bigger N is chosen, the smoother EMG_P will be at the cost of longer processing time. Here, we set N equal to 60 ms with the sampling frequency and signal characteristics into consideration.

The flow diagram of the proposed step detection algorithm is shown in Fig. 4(a). It is obviously found from the signal representation that for most locomotion patterns an activation segment is always located in the stance phase of a gait cycle. Although their time durations vary with walking or running speed, the time durations of EMG activation segments from muscle VL almost keep in a range (300–500 ms). In the following algorithm, we set the period of activation segments as a fixed value. If we can find a time interval in which the integral of signal values is a peak, then the activation segment can be detected, which are detailed as follows.

1) Apply a sliding window with the width of N epochs on the signal of EMG_P and get the integral signal $IEMG_P$

$$IEMG_P(i) = \sum_{j=i}^{i+N} EMG_P(j)/N. \tag{2}$$

The parameter N is set as 400 empirically in our experiments. Thus, $EMG_P(j)$ represents the energy of EMG in time j, and $IEMG_P(i)$ represents the integral of EMG energy between times i and $i+N$.

2) Taking advantage of the cyclic patterns of walking, a peak detection algorithm was performed on $IEMG_p$ to detect the step occurrence. The peak detection algorithm works as follows.

   a) First, initialize a sliding window with a proper width. The width of the sliding window is important for peak detection. False peaks will be detected when the window is too short. While, if it is set too long, some real peaks will be omitted. In our experiments, the width was set 600 epochs (equivalent with 600 ms) empirically to obtain high accuracy detection results.
   b) Second, check whether the biggest value in the sliding window locates in the middle of the window. If yes, and the value is bigger than the threshold value, then a peak (corresponding to the onset of a step occurrence) is detected.
   c) Finally, shift the sliding window one epoch forward and repeat the work until to the end of the signal sequence.

Fig. 4(b) gives an example of EMG signal processing procedures for step detection.

3) *Walking Pattern Recognition:* In the proposed EMG-based PDR solution, locomotion pattern recognition was an essential step that was performed prior to the final positioning. In a real life, common pedestrian's locomotion patterns include not only walking on the flat, but also walking with stairs and running. Apparently, it is difficult to classify the complicated daily locomotion patterns with signals from a single channel. Consequently, features from multichannel EMG signals were extracted and fed into a multistage decision tree for further locomotion pattern recognition.

a) *EMG features extraction:* Step frequency is an important parameter in gait analysis and shows a positive correlation with walking velocity, as well as amplitude level of EMG signals [30]. Since it was unpractical to calculate instantaneous step frequency, here we used an approximate parameter SF. SF can be obtained as the reciprocal of average interval time between two adjacent step occurrences.

In a stride cycle, different muscles contract coordinately to keep the body's balance, and the coordinate patterns differ from varying locomotion patterns. Specifically, the motion range of knee joint when walking on stairs is larger than that when walking on flat. Therefore, the amplitude of EMG signal from RF can serve as an indicator of classifying two situations. And the difference can be indicated by mean absolute value (MAV) of signals within activation segment. Also, when people step downstairs, their toes and heels contact with ground successively, which resulting two adjacent activation segments in one stride period of signals from VL. The step detection algorithm usually recognizes the start point of the second activation segment since the signals have bigger amplitudes. Thus, MAVs of signals within and before activation segments from VL are extracted as EMG features. MAVs are calculated by the formulas as follows:

$$MAV_{vl1}(i) = \frac{1}{N_1+1}\sum_{t=St(i)}^{St(i)+N_1} |s_{vl}(t)| \tag{3}$$

$$MAV_{vl2}(i) = \frac{1}{N_2+1}\sum_{t=St(i)-N_2}^{St(i)} |s_{vl}(t)| \tag{4}$$

$$MAV_{rf}(i) = \frac{1}{N_1+1}\sum_{t=St(i)}^{St(i)+N_1} \left|s_{rf}(t)\right| \tag{5}$$

where $s_{vl}(t)$, $s_{rf}(t)$ denote the EMG data of VL and RF respectively at time t, $St(i)$ is the ith step onset time (defined as the beginning of EMG activation segment from VL) we have detected. In our case $N_1 = 400$ ms (equivalent to the average length of activation segments), $N_2 = 200$ ms (equivalent to the average length of the first activation segments in one stride period when downstairs).

A pretest with four walking patterns was conducted to validate the effectiveness of classification using the extracted EMG features. Fig. 5 shows the step frequency and EMG MAV features of four different locomotion patterns. We can observe from Fig. 5(a) that SF of run is much larger than that of other motions, MAV_{vl1} and MAV_{vl2} are more sensible to upstairs and downstairs respectively, and MAV_{rf} shows varying levels with different motions. Fig. 5(b) demonstrates three motions are linear divisible with the EMG MAV features.

b) *Locomotion pattern recognition Algorithm:* A three-stage decision tree shown in Fig. 6(a) was applied to realize the classification of five kinds of locomotion patterns. At the first stage, standing still or standing with irregular lower limb motions was classified from regular walking behaviors with HMMs [30]. At the second stage, running was discriminated from walking upstairs, walking downstairs and walking on flat ground. The main point to distinguish walk and run is whether

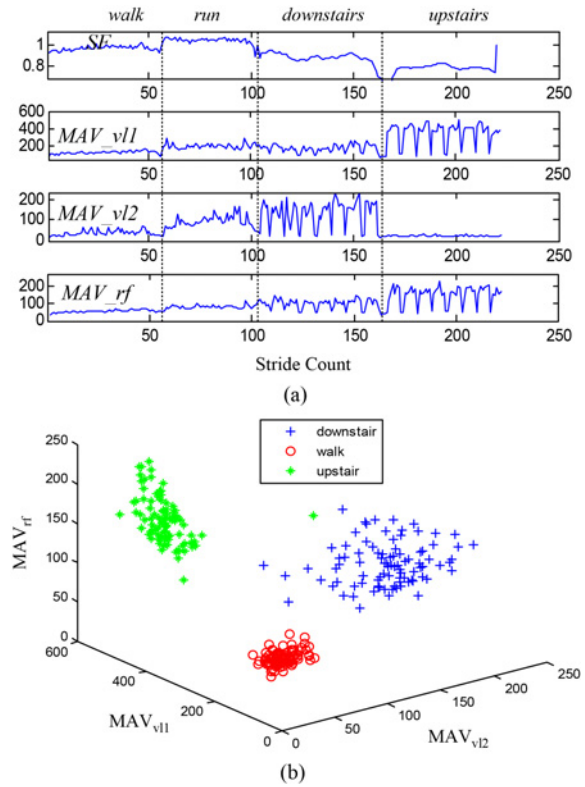


Fig. 5. (a) Step frequency and EMG features of different locomotion patterns. (b) Three motions are linear divisible with the EMG features displayed in (a).

there is a moment when both feet are away from the ground simultaneously in one stride period. For common people the step frequency of running is far bigger than that of walking, so running can be classified from other motion patterns by setting a proper threshold value of SF. As Fig. 5(b) shows, the rest three locomotion patterns are linear divisible with MAV features. Thus a 3-D vector formed by the MAV features was input to a linear discriminant classifier (LDC) [37] as shown in Fig. 6(b) to classify walking on the flat and up/downstairs at the third stage.

c) *Position calculation:* In order to calculate the pedestrian's location based on PDR algorithm, the information of two aspects is essential: 1) the stride length, and 2) heading angle. In this paper, a linear equation was utilized for stride length estimation, and the heading angles were accessed by fusing map information and outputs of digital compass embedded in our data collecting system.

4) *Stride Length Estimation:* Usually, the step length is defined as the distance between the heel on the right foot and the heel on the left foot when the pedestrian moves and takes a single step. And stride length equal to the distance from the heel of pedestrian's right foot to the heel of his right foot again after taking two full steps (left foot makes no difference here). Since stride length has a positive correlation [11], [30] with step frequency and certain EMG features, it is feasible to use a linear equation for stride length estimation.

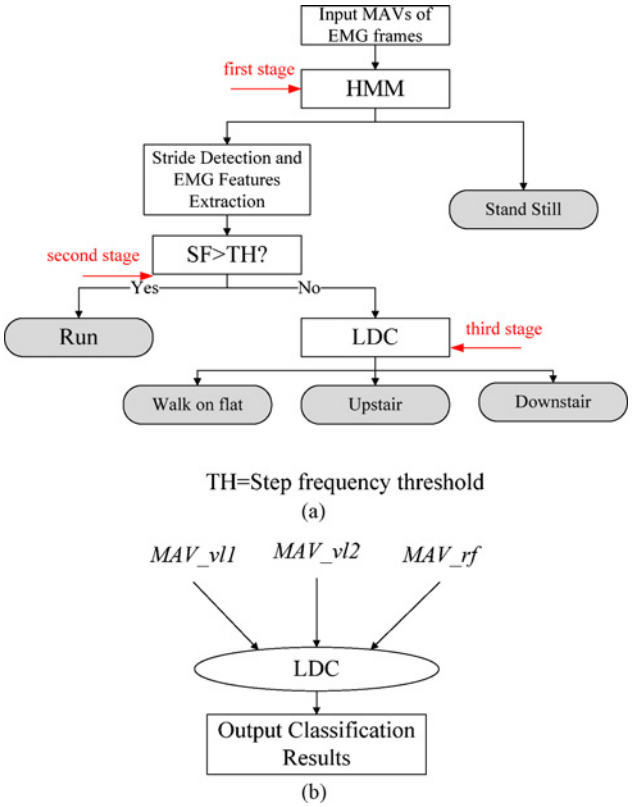


Fig. 6. (a) Diagram of three-stage decision tree for locomotion pattern recognition. (b) Diagram of linear discriminant classifier to classify walking on flat ground and up/downstairs, the inputs are EMG features extracted by (3)–(5).

The stride lengths for walking and running are modeled as follows:

$$SL(i) = a_0 + \sum_i a_i F_i^l + \sum_i b_i F_i^r \tag{6}$$

where *SL(i)* denotes the *i*th stride length, F_i^l and F_i^r are *i*th EMG features of left and right leg respectively, and a_i,b_i are the estimated coefficients that can be determined by a least-squares method during the training phase. The EMG features taken into consideration for stride length equations include *SF*, *MAVs* mentioned above and mean absolute difference value (MADV) that can be calculated by

$$MADV_{vl1}(i) = \frac{1}{N_1 + 1} \sum_{t=St(i)}^{St(i)+N_1} |s_{vl}(t+1) - s_{vl}(t)|. \tag{7}$$

In the case of stairs situation, we suppose that each stage of stair has a fixed width and height, which is reasonable for the same building in context-awareness and healthcare applications, so *SL* in this case equals stair's width.

5) *Dead Reckoning:* A digital compass senses the geomagnetic field and gives the angle of deviation between walking direction and the magnetic north. However, a low cost 2-D digital compass always suffers from disturbances derived from environment and system setup inclination. In this paper, map matching technology, which assumes the possible heading angles make up a finite set: $\{\alpha_1, \alpha_2, \ldots, \alpha n\}$, and determines the accurate heading by choosing the proximal element with the output of digital compass, was applied to improve the heading accuracy. Since the probable travel directions indoor are restricted by the terrains or building structures and outdoor walking is more casual without many restrictions, map matching technology is more suitable for indoor environments. Considering our special application (indoor positioning) scenarios [35], map matching technology is effective and available to acquire accurate heading information.

Combined with the estimated stride lengths, positions can be reckoned with formulas as follows:

$$\begin{cases} N_{k+1} = N_k + SL_k \cdot \cos\alpha_k \\ E_{k+1} = E_k + SL_k \cdot \sin\alpha_k \\ A_{k+1} = A_k + SA_k \end{cases} \tag{8}$$

where N_k and E_k are the coordinates of the North and the East, which represent position when the *k*th step occurs, A_k are the attitude on the vertical direction. SL_k and α_k are the corresponding stride length and azimuth of the *k*th step, SA_k are the rise on vertical direction of the *k*th step, which should be equal to the height of the stage.

III. Experiments and Results

A. Experiments

The tests were carried out in the west campus of University of Science and Technology of China (USTC). We designed several experiments to verify the efficiency of the EMG-based PDR method proposed in this paper. The first stage experiment is to validate the efficiency of locomotion pattern recognition method, and the second stage contains two field tests for the proposed EMG-based PDR solution. The first stage's data were also used for training the LDC and stride length equations that were used by second stage's field tests.

The first stage experiment was carried out in an experimental building in the university campus. A 22-year-old male subject with the wearable signal collection system as shown in Fig. 7 walked in four patterns to collect signals: process 1) walking along the corridor from one end to the other, the distance measuring about 76.2 m long; process 2) running along the corridor from one end to the other; process 3) walking downstairs from fourth flour to the basement, about 80 stair steps in all; and process 4) walking upstairs back to the fourth flour. Among them, process 1) was repeated for nine times, with three times for slow, normal, fast velocity each; process 2) was repeated for ten times, with twice for five various velocities each; and the other two processes were all repeated three times in our first stage experiment.

The second stage experiment contained two field tests. One is on the playground, and the same subject in the first stage experiment walked along the first track of the standard sports field with a perimeter of about 404 m for three times. From the start point, the subject experienced slow walk, normal speed walk, fast walk successively, and returned to the start point finally. The same tester walked with random speeds from an outdoor start point, entered into the building, climbed to the third floor, went through the corridor, and went downstairs to the first floor to simulate a seamless

Fig. 7. Field test environments.

TABLE I
RECOGNITION RESULT WITH 20 SAMPLES FOR TRAINING, 70 FOR TESTING

Locomotion pattern	Running	Walking	Upstairs	Downstairs	Recognition rate(%)
Running	70	0	0	0	100
Walking	0	70	0	0	100
Upstairs	0	0	69	1	98.57
Downstairs	0	0	0	70	100

indoor-outdoor positioning scenario in the fourth electricity building. The experiment scenes are shown in Fig. 7.

B. Step Detection and Locomotion Pattern Recognition

The step detection algorithm proposed in this paper was applied to the experiment data, and obtained good results with the average misdetection rate less than 1%. Furthermore, the algorithm is robust for all the locomotion patterns concerned in this paper. Misdetection is mainly caused by signal noise or irregular motions like sharp stops when running. This kind of misdetection can be reduced by improving the sensor properties and adjusting sensor positions.

To validate the recognition method, 90 steps for each motion from first stage's experiment were used in the locomotion pattern recognition experiment. Four EMG features (SF, MAV_{vl1}, MAV_{vl2} and MAV_{rf}) mentioned in part II were extracted as input to the three-stage decision tree. The threshold value of step frequency (TH) in the second stage decision is set to be 1.25 so that every step in running can be recognized. The LDC model was trained with part of samples from the 90 steps, and the remaining samples are used for testing. As Table I shows, experiment results show good performance when we used only 20 samples for training, and the worst recognition rate is more than 98%.

C. Stride Length Estimation

Nine groups of walking tests and ten groups of running tests conducted in the first stage experiment are used firstly for EMG features selection in stride length equations. For each test group, we calculated the average stride length (SL) and average

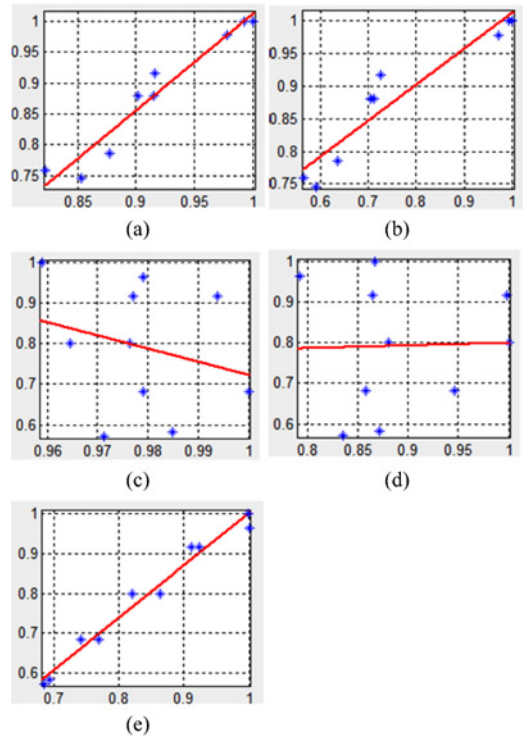


Fig. 8. Relations between stride length and EMG features of walking and running patterns.

EMG features (SF, MAV_{vl1} or $MADV_{vl1}$) of all the strides. Then, the SLs and EMG features of nine walking groups and ten running groups are normalized to 0∼1, respectively.

Fig. 8 shows the relationship between SL and SF, MAV_{vl1} or $MADV_{vl1}$ for walking and running respectively. We found that SF and MAV feature from VL are approximately linear correlated with the real SL when walking (Fig. 8(a) $R^2 = 0.938$, and (b) $R^2 = 0.903$). So, a linear stride length equation is defined as (9) for walking

$$SL(i) = A \bullet SF(i) + B \bullet LMAV_{vl1}(i) + C \bullet RMAV_{vl1}(i) + D \quad (9)$$

where $LMAV$ and $RMAV$ denote the EMG features from left and right legs, respectively.

However, (9) cannot be used to estimate stride length for running situations, since stride length keeps uncertain correlation with step frequency and MAV_{vl1} [Fig. 8(c) and (d)]. So, we alter (9) to the form as (10) with $MADV_{vl1}$ features (from both left and right legs) that are linear correlation with the running SL (Fig. 8(e), R^2=0.975)

$$SL(i) = A \bullet LMADV_{vl1}(i) + B \bullet RMADV_{vl1}(i) + C. \quad (10)$$

A cross-validation experiment was conducted to investigate the effectiveness of the stride length estimation method proposed above. Six groups of data from process 1) were used for training the linear stride length equation and the remaining three groups for testing. To obtain better results, we made sure that in each training phase, the number of samples with slow, normal and fast velocities should be equal. Average stride length (AvSL) calculated by dividing the total distance by steps

TABLE II
STRIDE LENGTH ESTIMATION RESULTS OF WALK AND RUN

Test Group	AvSL (m)	Estimated (m)	Error (%)
TGW01	1.3487	1.4522	7.67
TGW02	1.2807	1.3585	6.07
TGW03	1.3026	1.2761	2.03
TGW04	1.5089	1.4796	1.94
TGW05	1.5711	1.4873	5.33
TGW06	1.5089	1.4715	2.48
TGW07	1.7124	1.7092	0.19
TGW08	1.6933	1.6659	1.62
TGW09	1.7125	1.7096	0.17
Average	**1.5155**	**1.5122**	**3.06**
TGR01	1.8585	1.8474	0.59
TGR02	1.8585	1.8459	0.67
TGR03	1.5551	1.6071	3.34
TGR04	1.5875	1.6285	2.58
TGR05	2.1771	2.2622	3.91
TGR06	2.1771	2.0793	4.49
TGR07	2.4984	2.4549	1.74
TGR08	2.4984	2.3852	4.53
TGR09	2.7214	2.8152	3.45
TGR10	2.6276	2.7410	4.32
Average	**2.1560**	**2.1667**	**2.96**

number was used as an approximate real value since the tester keeps the same cadence in the process of each experiment stage.

Similarly, (10) was validated by another ten groups of running data. Table II gives the stride length estimation results with (9) and (10), where TGW01–TGW09 denote walking groups, and TGR01–TGR10 denote groups of running. The average stride length accuracy of both walking and running is about 97% with the worst situation less than 93%. Bad outcomes of some groups like TGW01, TGW02 and TGW05 may result from kinds of factors, such as the move of sensor's locations which affect the signals' amplitudes a lot. Also, the walking velocities in our experiments extend to a very large range to cover as many situations as possible.

D. Field Tests Result

Algorithms proposed in this paper also demonstrated good performance in our field tests. Stride length equation and LDC used here were trained by data from first stage's experiments. To get the best result, we selectively used six groups (TGW01, 02, 04, 05, 07, and 08) of walking data for training. Parts of our experiments were conducted outdoor, and the routes of the subject were designed beforehand. So, heading information used in dead reckoning was obtained with the fusion of map information and outputs of low-cost digital compass. Trajectories of GPS solutions and the true trajectories from Google Earth are used as comparisons. Due to the precision of GPS results, real-time deviations were unavailable, so the total estimated distance and the real distance is given below.

1) *2-D Environment Test:* The 2-D environment tests were conducted on the playground with an open sky, thus effective GPS signals can be used as a comparison of our positioning results. Locomotion pattern in the tests is walking in various speeds. The results of the three tests in the second stage on the playground are listed in Table III. Rd01∼03 represents the first to third test around the playground. The error of stride length estimation ranges from 1.33% to 2.17%.

TABLE III
STEP DETECTION AND DISTANCE ESTIMATION RESULTS OF FIELD TESTS ON PLAYGROUND

Test Group	Real Steps	Detected Steps/error	Real Distance(m)	Estimated Distance(m)/error
Rd01	282	282 / 0	404	395.20 / 2.17
Rd02	268	268 / 0	404	397.39 / 1.64
Rd03	265	265 / 0	404	409.36 / 1.33

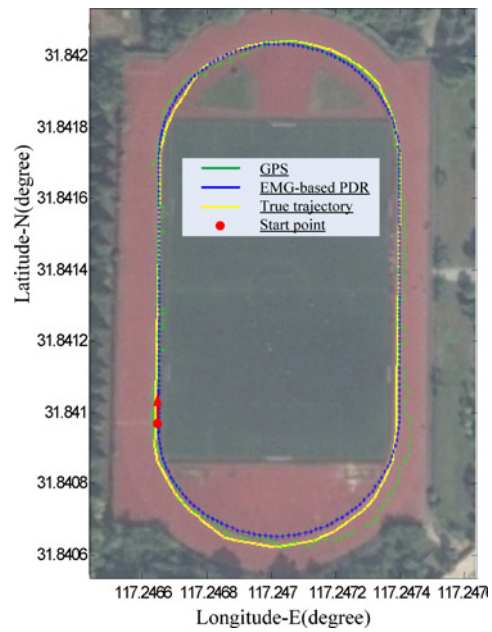


Fig. 9. Trajectory of EMG-based PDR versus the reference trajectory of GPS solution in the playground field test.

Fig. 9 demonstrates the PDR result of experiment Rd03. The EMG-based PDR showed a comparable result with a low-cost commercial GPS module. In the period of about 300 s, which is equivalent to a walking distance of 404 m, the estimated distance was 409.36 m, with an error rate of less than 2%. The technology of map matching eliminated the error caused by inaccurate heading, so the deviation mostly came from stride length estimation. The deviation was accumulated over time, and the accumulated distance between real and calculated position was added up to about 5 m. The positioning accuracy showed an apparent improvement compared with our pilot work.

2) *3-D Environment Test:* In order to extend the EMG-based PDR solution to a 3-D environment application, we designed an experiment to simulate a common situation in which people walks from outdoor to indoor, and the GPS signal degraded quickly. A motion recognition algorithm was applied to recognize the locomotion patterns, and among the entire 323 steps, only one walking step was identified as downstairs falsely. So, the recognition rate was more than

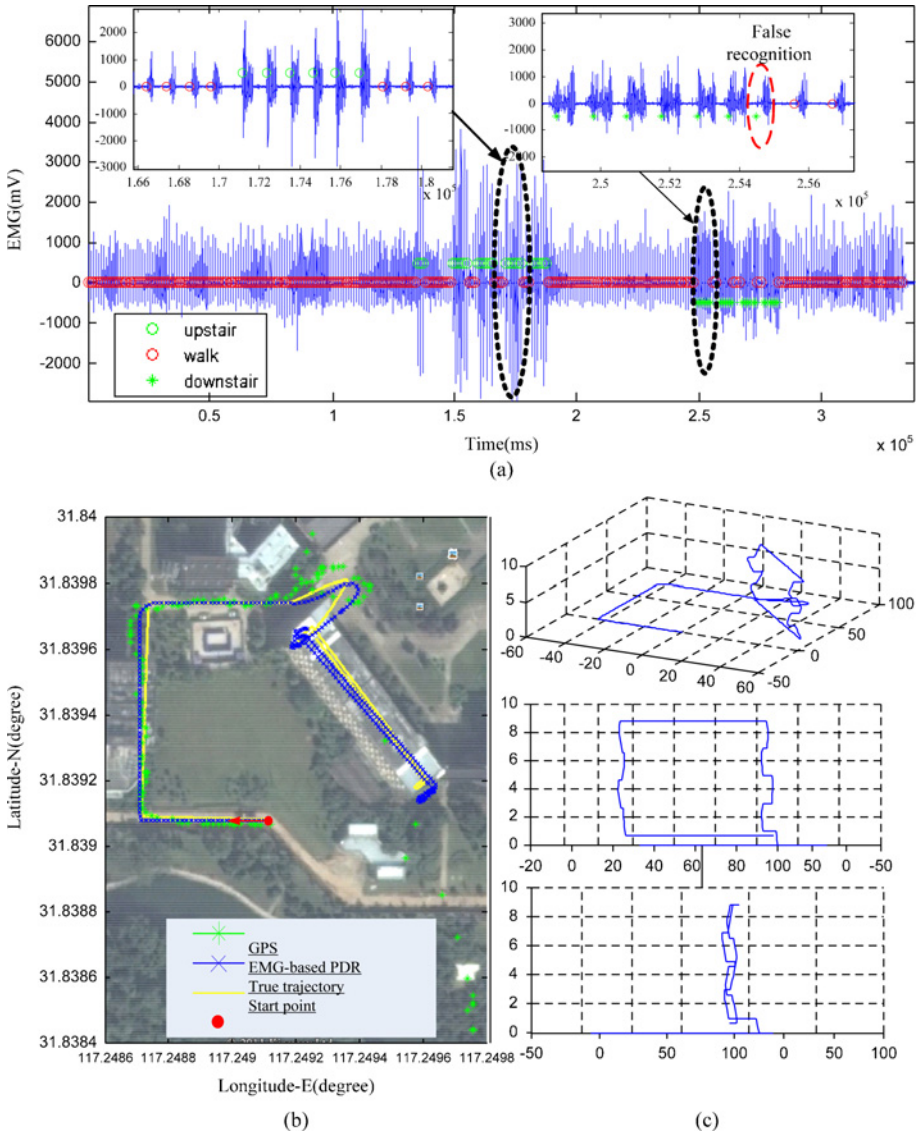


Fig. 10. EMG-based PDR in 3-D environment. (a) Locomotion pattern recognition results marked, respectively, on the raw EMG signals. (b) Trajectory of EMG-based PDR versus GPS solution in the out/indoor field test. (c) Trajectory of EMG-based PDR from three different views (general view, from the front face of the building, from the side face of the building).

99% and the deviation in the vertical direction was one stair's height that equals 0.163 m in our case. The deviation rate in the horizontal plane was comparable with that in the 2-D environment test. The real and estimated total distances were 362 and 355.21 m, respectively. The biggest deviation came before entering the building because of the ineffective map matching technology. Fig. 10 shows the locomotion recognition results and PDR results from different views. We can see that in outdoor environment, the EMG-based PDR solution produced a comparative trajectory with the GPS solution. However, when the subject went into the building, the GPS solution offered poorer position accuracy, while the EMG-based PDR solution offers more credible results. Thus, as an alternative indoor positioning method, the EMG-based PDR solution was more effective in indoor/outdoor seamless positioning applications.

IV. Conclusion and Future Work

In this paper, we proposed an EMG-based PDR solution toward improved context awareness applications. In the proposed method, multichannel EMG signals recorded from lower limb muscles were used to identify pedestrian's activities first, and then different stride length models were developed for four locomotion patterns according to their kinesiological properties. Furthermore, considering the error of heading information derived from low-cost digital compass, map matching technology was applied to reduce the position deviation resulted from heading errors. A series of experiments were conducted to validate the effect of the proposed method in locomotion pattern recognition and personal positioning. The experimental results demonstrated that EMG-based pattern recognition of four motions (walking, running, walking upstairs, walking downstairs) achieved an error rate of less than

2%, and the proposed EMG-based PDR solution obtained a position deviation of less than 5 m within the whole distance of 404 m in a simulated indoor/outdoor field test.

Although we researched an effective EMG-based PDR solution to indoor personal positioning that enables potential context-awareness applications and mobile healthcare, there are still some limitations that need to be further studied.

Research using EMG signals to sense the human body movements has been prevalent in hand gesture recognition, human–computer interaction [28], and rehabilitation [23], [24], [39]. In this paper, multichannel EMG signals were collected from lower extremity to classify pedestrian's activities toward context-awareness applications. As a preliminary study, we considered five locomotion patterns and used two EMG sensors to collect signals from each thigh, and obtained the satisfying user-dependent classification results. Activity patterns considered are mainly concerned with pedestrian's locomotion, and muscles accounting for these locomotions distribute from hip down to foot. Without a doubt, the more EMG sensors are used, the lower classification error rate will be obtained. Also, combinations of different muscle groups can generate different classification effects concerning the same locomotion pattern set [39]. In addition, the EMG signals vary a lot from subject to subject, which makes it a problem to realize user-independent stride length estimation. Proper amplitude normalization of EMG signals, which can provide a way to eliminate the differences among different subjects, will be explored in future algorithm optimization.

EMG signals response not only to obvious body motions, but also some unapparent movements, even some mental activities without physical manifestations. EMG signals also contain much useful information (e.g., fatigue extent and muscular disease assessment). So, EMG signals will play an important role in healthcare applications for the elders and patients. In addition, the walking locomotion pattern and pedestrian position considered in this paper, EMG signal, as a kind of physiological parameter, has a potential value in sensing more complex human contexts. Combined with other physiological parameters (e.g., pulse, oxyhemoglobin saturation, and skin resistance), and mechanical parameters (e.g., acceleration and speed), we can try to sense the context more intelligently in the future.

High-accuracy heading sensors are restricted in practical applications for the cost, while low-cost sensors suffer large error from the environments. Although heading error can be reduced by some algorithms, the methods are usually complex and inconvenient. Because indoor environments are mainly places such as apartment, office building, exhibiting hall, etc., the technology of map matching [8], [35] is effective as an auxiliary method to improve the accuracy of heading. In addition, map information of the environment can also be used to verify and rectify the pattern recognition results and positions in our future work. For example, the locomotion pattern is restricted only to walking up/downstairs when the person is on a staircase and the position of staircase on maps can be served as a reference point for the EMG-based PDR solution to rectify the real-time position. Although we introduced map matching technology to reduce the heading error and generated satisfying results, the algorithm is noneffective in the outdoor or open environments. Part of the restrictions can be reduced by using more intelligent data fusion algorithms, such as particle filter [40] and Kalman filter [35] in the future work.

Finally, a real-time system with multisensor signals collection and analysis is expected for the context-awareness applications such as mobile healthcare for the elders and patients. Although some locomotion patterns are not yet included in our current experiments, a more complex and general motion set will be considered in our future work including slope, jump, sit down, fall over, etc., to apply our algorithm to other circumstances with complex terrains. In addition, improvements should also be made in the following aspects: 1) wireless sensors are necessary to guarantee comfortable user experience; 2) algorithms should be improved to satisfy multiuser and multiday applications; and 3) data fusion technology should be used to analyze multisource information.

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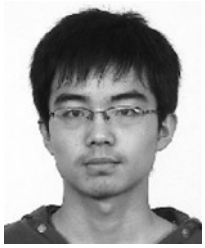

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